

Praxia Update

Daily orthopaedic literature digest · Friday, 12 June 2026 · 57 new articles across 10 journals



American Journal of Sports Medicine

June 2026 · online ahead of print · 4 new articles

Partial Tenotomy and Reattachment for Chronic Proximal Hamstring Tendinopathy: Functional Outcomes, Return to Sport, and Patient Satisfaction.

Lefèvre N, Moussa MK, Valentin E, Hadjam T, Gerometta A, Grimaud O, Meyer A, Bohu Y et al.

BACKGROUND: Chronic proximal hamstring tendinopathy (PHT) can cause significant pain and functional impairment, particularly in active individuals. While nonoperative management is the first-line treatment, some patients with persistent symptoms may require surgical intervention.

PURPOSE: To evaluate the functional outcomes, return-to-sport rates, and complication profiles of patients undergoing partial tenotomy and reinsertion for chronic PHT.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This retrospective cohort study targeted patients who underwent partial tenotomy and reattachment of the proximal hamstring tendons between April 2010 and September 2022. The indication for surgery was refractory chronic PHT symptoms for >6 months. The primary outcome measure was the Parisian Hamstring Avulsion Score (PHAS). Secondary outcomes included the Tegner Activity Scale score, University of California-Los Angeles (UCLA) Activity Scale score, return to sport, postoperative patient satisfaction, and the rate of complications. In this study, the terms "partial involvement" or "complete involvement" refer to the extent of magnetic resonance imaging signal changes across the tendon, rather than to true tendon ruptures or avulsions.

RESULTS: The study included 42 patients with a mean age of 50.4 years (SD, 10.4 years). The mean follow-up was 4.1 years (range, 2.0-12.6 years [...])

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Revascularization of Transcortical Vessels Improves Tendon-to-Bone Healing for Rotator Cuff Repair.

Xia W, Shi Z, Wang X, Wu H, Lu J, Yu X, Shan H, Deng C et al.

BACKGROUND: Rotator cuff tear is a prevalent musculoskeletal disorder with a high postoperative retear rate. Transcortical vessels (TCVs) are capillaries that cross cortical bone and represent key elements of bone microcirculation. However, the possible intervention and influence of TCVs for rotator cuff repair (RCR) have rarely been studied.

PURPOSE: To investigate the effect of TCV-mediated revascularization in RCR and the mechanism of potential intervention.

STUDY DESIGN: Controlled laboratory study.

METHODS: Sprague-Dawley rats (N = 135) with supraspinatus tendon tear and delayed repair were randomly assigned into 3 groups based on footprint preparation: (1) cortical bone freshening up by debridement to expose cancellous bone, (2) cortical bone preservation as the control group, and (3) cortical bone preservation with oral D-sphingosine as a TCV-stimulating factor. Oxidative stress was measured using fluorescent probes; hypoxia and inflammation levels were measured via immunosorbent assay; and RCR and TCVs were evaluated with biomechanical and immunostaining assays.

RESULTS: Cortical bone preservation inhibited the revascularization of TCVs during RCR ($P < .01$), while freshening up or D-sphingosine treatment promoted TCV formation by reducing oxidative stress ($P < .05$ for all). Revascularization of TCVs alleviated hypoxia and inflammation-derived chronic tendinopathy af [...]

The Effect of Preoperative Symptom Duration on 10-Year Outcomes After Hip Arthroscopy for Femoroacetabular Impingement Syndrome: A Matched Analysis.

Hu EY, Bi AS, Cervantes JE, Nho SJ

BACKGROUND: Patients with femoroacetabular impingement syndrome (FAIS) experiencing >2 years of pain before hip arthroscopy have been linked with worse short-term and midterm outcomes.

PURPOSE: To examine the effect of preoperative pain duration on patient-reported outcomes (PROs), clinically significant outcomes, and reoperation rates in patients undergoing primary hip arthroscopy for FAIS.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A prospectively maintained surgical repository was reviewed to select patients who underwent primary hip arthroscopy for FAIS between January 2012 and October 2014 with 10-year follow-up. Patients who reported pain ≥ 2 years before surgery were propensity score matched 1:1 to patients reporting preoperative pain <2 years by age, sex, and body mass index (BMI). PRO scores collected included those for the Hip Outcome Score-Activities of Daily Living (HOS-ADL), Hip Outcome Score-Sports Subscale (HOS-SS), modified Harris Hip Score (mHHS), and visual analog scale (VAS) for pain and satisfaction. Achievement rates of the minimal clinically important difference and patient acceptable symptom state were compared. Reoperation-free survivorship was compared with Kaplan-Meier analysis.

RESULTS: A total of 164 patients who had a symptom duration ≥ 2 years (mean age, 35.9 ± 11.7 years; 64.0% female; mean BMI, 25.3 ± 5.2 kg/m²) were successful [...]

Contralateral ACL Rupture as an Indicator of Intrinsic Risk for Graft Failure After ACL Reconstruction: A Cohort Study of 7718 Patients From the SANTI Study Group.

Santamaria F, Culebras Almeida LA, Carrozzo A, Campos JP, Saithna A, Vieira TD, Sonnery-Cottet B

BACKGROUND: Graft failure after anterior cruciate ligament (ACL) reconstruction (ACLR) may be influenced by both extrinsic exposure and intrinsic patient susceptibility. A contralateral ACL rupture could reflect an inherent predisposition to ligament injury.

PURPOSE: To determine whether a history of bilateral native ACL ruptures increases the risk of graft failure after primary ACLR.

STUDY DESIGN: Cohort study; Level of evidence, 2.

METHODS: A total of 7718 consecutive patients who underwent primary ACLR by a single surgeon (2003-2022) were included. Graft failure, defined as a clinically and magnetic resonance imaging-confirmed rupture, was compared between patients with bilateral versus unilateral native ACL ruptures. Multivariable logistic regression identified independent predictors, including age, sex, Tegner score, sport type, and a history of bilateral rupture.

RESULTS: The cohort included 6327 patients with isolated native ACL rupture and 1391 patients with bilateral native ACL rupture. The overall graft failure rate was 6% (461-7718), with a mean follow-up of 135.9 months. Patients with bilateral ACL ruptures had a higher failure rate (9.3%) than those with unilateral injuries (5.2%) ($P < .0001$). Independent predictors included younger age (odds ratio [OR], 0.92 [95% CI, 0.90-0.93]; $P < .0001$), male sex (OR, 1.46 [95% CI, 1.10-1.96]; $P = .0089$), higher Tegner score [...]

Does Liposomal Bupivacaine Provide Superior Postoperative Analgesia Compared With Ropivacaine With Dexamethasone in Suprainguinal Fascia Iliaca Compartment Block for THA? A Randomized Controlled Trial.

Wu W, Hu X, Yang C, Sun Y, Li J, Wang L

BACKGROUND: Choosing the most effective pain management is important for enhancing functional recovery after surgery. The fascia iliaca compartment block (FICB) is recommended for analgesia after THA, but whether liposomal bupivacaine is more effective for this application remains uncertain.

QUESTIONS/PURPOSES: (1) Did a single-injection suprainguinal FICB using liposomal bupivacaine provide superior pain control by a clinically important margin? (2) What was the effect of liposomal bupivacaine on perioperative opioid consumption? (3) How did liposomal bupivacaine affect the duration of the sensory block and quadriceps motor weakness in the operated limb?

METHODS: This was a prospective, parallel-group, assessor-blinded, single-center, randomized controlled clinical trial. Patients who underwent unilateral primary THA via the posterolateral approach were included in the study. Exclusion criteria included a BMI of > 32.5 kg/m² (which is the diagnostic criterion for moderate obesity in China), contraindications to peripheral nerve block, known allergy to study medications, psychiatric or cognitive disorders that could interfere with pain assessment, severe hepatic or renal dysfunction, and chronic pain necessitating long-term opioid therapy. According to these criteria, 76% (60 of 79) of screened patients met eligibility requirements and were subsequently randomized into either [...]

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Is Neighborhood Socioeconomic Deprivation Associated With Outcomes Following Primary THA and TKA? A Systematic Review Utilizing the Area Deprivation Index.

Furyes AR, Elmenawi KA, Hackett L, Deren ME, Piuze NS

BACKGROUND: Socioeconomic disadvantage is increasingly recognized as an important factor associated with outcomes after total joint arthroplasty (TJA). The Area Deprivation Index (ADI) has emerged as a validated measure of socioeconomic disadvantage that specifically captures individual neighborhood levels of structural and material deprivation based on income, education, employment, and housing characteristics. Prior research has demonstrated ADI to be associated with surgical outcomes; however, its relationship to postoperative outcomes after THA and TKA, procedures in which outcomes depend highly on institutional and structural support, home environment, and accessible postacute care resources, has not been well synthesized. Prior studies provide a broad overview of heterogeneous measures of socioeconomic disadvantage that are not directly comparable.

QUESTIONS/PURPOSES: (1) Is a higher ADI associated with increased postoperative healthcare utilization after primary THA and TKA? (2) Is a higher ADI associated with differences in patient-reported outcome measures (PROMs)? (3) Is a higher ADI associated with increased postoperative complications? (4) Is ADI associated with differences in implant survivorship after TJA?

METHODS: A systematic review of four databases-Ovid Medline, Embase, Cochrane Library, and Web of Science Core Collection-was performed from their inception to [...]

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Editor's Spotlight/Take 5: How Does Traumatic Hand Injury Impact Patients in a Safety-net Healthcare System? A Mixed Methods Study.

Wongworawat MD

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Letter to the Editor: Guest Editorial: Recalling a Recall.

Doyle TR, Hennessy O, Keogh CJ, Hughes AJ, Kenny PJ, Sheridan GA, Murphy CG

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CORR Insights®: Is Higher Surgeon Volume Associated With Lower Complication and Revision Risk After Distal Radius Fracture Surgery? A Population-based Cohort Study of 13,389 Patients.

O'Shaughnessy MA

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Is Higher Surgeon Volume Associated With Lower Complication and Revision Risk After Distal Radius Fracture Surgery? A Population-based Cohort Study of 13,389 Patients.

Persitz J, Zagorski B, Baltzer H, Calzavara A, Wolfstadt J, Avisar E, Zuo KJ, Chan A et al.

BACKGROUND: Surgeon case volume has been linked with outcomes across many orthopaedic procedures, but its influence on distal radius fracture fixation remains uncertain.

QUESTIONS/PURPOSES: (1) For distal radius fracture surgery, at what surgeon annual case volume does the risk of complications plateau? (2) For distal radius fracture surgery, at what surgeon annual case volume does the risk of revision surgery plateau?

METHODS: A retrospective, population-based study was performed using administrative health databases in Ontario, Canada, accessed through the Institute for Clinical Evaluative Sciences, an independent, nonprofit research institute that houses linkable, individual-level health administrative data for Ontario's publicly funded healthcare system. Between 2010 and 2020, a total of 27,945 adult patients (≥ 18 years of age) underwent surgical fixation for acute isolated distal radius fracture. After applying prespecified inclusion and exclusion criteria, including exclusion of patients with open fractures, polytrauma, compartment syndrome, neurovascular injury, emergent presentations, incomplete administrative records, or prior distal radius surgery, a final cohort of 13,389 patients (48% of the initial cohort) was included (71% [9533] females; mean $\pm$ SD age 56 ± 15 years). Surgeon annual case volume, defined as the number of distal radius fracture fixations performed [...]

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On Patient Safety: Worsening Healthcare Costs Threaten Americans' Safety.

Rickert J

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Comparative Analysis of Immune Cell-Type Abundances in Periprosthetic Tissues Across Arthroplasty Failure Etiologies: Use of Transcriptomic Deconvolution.

Li Y, Wang F, Li Y, Mu W, Ji B, Zhang X, Cao L

BACKGROUND: Although cellularity is traditionally evaluated morphologically, an emerging transcriptome-sequencing-based algorithm enables simultaneous inference of cellular information. We studied whether cellularity profiles predicted using CIBERSORTx would (1) depict immune cell-type abundances in periprosthetic tissues across arthroplasty failure etiologies, and (2) provide prognostic value for identifying cases of periprosthetic joint infection (PJI).

METHODS: CIBERSORTx-derived cellularity profiles were evaluated in 185 periprosthetic tissue samples, including 135 from patients with PJI (64 males; median age, 66 years) and 50 from those with aseptic failure (AF) (36 males; median age, 62.5 years), that had been subjected to bulk RNA sequencing. Kaplan-Meier survival analysis was performed to assess prognostic outcomes in PJI.

RESULTS: Of the 22 evaluated cell types, 5 were significantly elevated in PJI cases: plasma cells, resting memory CD4+ T cells, CD8+ T cells, activated mast cells, and M1 macrophages (all $p < 0.05$ after Benjamini-Hochberg [BH] correction). Conversely, 3 cell types were significantly elevated in AF cases: gamma delta T cells, M0 macrophages, and M2 macrophages (all $p < 0.05$ after BH correction). Of the combined immune cell populations, total B cells, total T cells, and natural killer cells were significantly elevated in PJI cases, while total macrophag [...]

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Minimal Movement Restrictions Do Not Increase Hip Dislocations Following Total Hip Arthroplasty: A Before-and-After Study of 10,357 Patients.

Klaassen AD, Willigenburg NW, Musters JWQ, Jager J, Poolman RW, Santeon Hip Osteoarthritis Group

BACKGROUND: Hip dislocation is the most common reason for early revision after total hip arthroplasty (THA). To prevent dislocation, patients are often prescribed movement restrictions following THA; however, these restrictions can be uncomfortable and limiting for patients and may delay a return to daily activities. The present study aimed to investigate whether a protocol with reduced movement restrictions following THA would lead to an increased incidence of early hip dislocations.

METHODS: In this uncontrolled, multicenter, before-and-after study, 2 groups were retrospectively compared. Patients who underwent a THA between January 1, 2015, and March 31, 2020, were included. Patients in the first group ($n = 7,666$) were prescribed strict movement restrictions following THA, and patients in the second group ($n = 2,691$) were prescribed minimal movement restrictions. Patient and prosthesis characteristics were collected, as well as data regarding hip dislocations within 90 days postoperatively as the primary outcome.

RESULTS: The incidence of early hip dislocations was not significantly different between movement groups, with 112 (1.46%) of 7,661 hips in the restricted group and 52 (1.93%) of 2,691 hips in the minimally restricted group experiencing a dislocation ($p = 0.093$).

CONCLUSIONS: The use of a protocol with minimal movement restrictions following THA did not increase [...]

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With Gratitude to Dr. Thomas Einhorn, Founding Editor of JBJS Reviews: Honoring a Legacy of Orthopaedic and Editorial Leadership.

Bhandari M, Swiontkowski MF

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A Proposed Alternative Sampling Strategy for Centers for Medicare & Medicaid Services Arthroplasty Patient-Reported Outcome Performance Measure Compliance.

Lyman S, Chin A, Alschuler DM, Brill CO, Fontana M, MacLean C, McLawhorn A

INTRODUCTION: The Centers for Medicare & Medicaid Services (CMS) mandate reporting of Hip disability and Osteoarthritis Outcome Score for Joint Replacement (HOOS, JR.) and Knee injury and Osteoarthritis Outcome Score for Joint Replacement (KOOS, JR.) to support a hospital-level patient-reported outcome performance measure (PRO-PM). The current CMS requires matched pre- and postoperative PROMs for $\geq 50.0\%$ of all eligible patients, regardless of hospital volume. This may yield unreliable performance estimates for low-volume centers and potentially biased and unnecessarily large estimates for large hospitals. The purpose of this study was to determine the smallest sample required to produce statistically valid PRO-PM estimates.

METHODS: We calculated the minimum number of completed PROMs needed in order to achieve a 95% confidence interval (CI) width 10.0% ($\pm 5.0\%$) for hospitals with five, 50, 500, and 5,000 eligible total hip arthroplasty (THA) and total knee arthroplasty (TKA) cases annually, assuming a conservative substantial clinical benefit (SCB) achievement of 50.0%. These values were compared with current CMS requirements. Sensitivity analyses explored the effects of confidence interval width, hospital volume, and frequency of SCB attainment on required sample sizes.

RESULTS: The CMS's volume-proportional rule requires three, 25, 250, and 2,500 patients for hospitals wit [...]

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First Revision Total Hip Arthroplasty for Instability: 10-Year Outcomes and Risk Factors Associated with Re-Revision.

Yapp LZ, Howard LC, Greidanus NV, Garbuz DS, Masri BA, Neufeld ME

BACKGROUND: Recurrent instability remains a leading cause of failure after revision total hip arthroplasty (rTHA). This study assessed the 10-year survivorship of the first rTHA when undertaken for instability and identified factors associated with failure.

METHODS: We retrospectively identified all consecutive rTHA performed for instability at our institution between 2000 and 2022. Of the 678 rTHA identified, 284 (41.9%) were the first rTHA and included in the study. Re-revisions and oncology cases were excluded. The mean age at revision was 65 years, and 63.2% were women. The mean follow-up was 11.4 years (range, two to 22). A total of 84 (29.6%) hips underwent at least one re-revision. Survival free from instability-related and all-cause re-revision at 10 years was calculated using Kaplan-Meier analysis. Multivariable Cox proportional hazard regression modeling was used to assess factors associated with 10-year survival.

RESULTS: The 10-year survivorship was 82.2% for instability and 73.6% for all-cause re-revision. Dual mobility (DM) components had higher 10-year instability-free survival than both standard bearings and constrained liners ($P = 0.04$). However, when considering 'large' (≥ 36 mm) femoral heads, there was no difference in the revision rate when compared to DM components ($P = 0.59$). Constrained liners were associated with a higher risk of revision for instabil [...]

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Jumbo Femoral Heads in Total Hip Arthroplasty: Improved Early Stability Without Added Complications.

Kim BI, Cheng R, Grabov E, Debbi EM, Vigdorichik JM, Mayman DJ, Shen TS

BACKGROUND: Larger femoral heads have been proposed to reduce dislocation risk following total hip arthroplasty (THA). While the benefit of increasing head size from 28 or 32 mm to ≥ 36 mm is well established, it remains unclear whether jumbo (40 or 44 mm) femoral heads confer additional stability over traditional large (36 mm) heads. This study aimed to determine whether jumbo heads improve stability or alter early complication rates.

METHODS: We identified 6,413 primary THAs with large-diameter femoral heads (36-, 40-, or 44-mm) implanted at a single institution between February 1, 2016, and December 31, 2022. Cases were stratified by femoral head diameter into the traditional large-head (36 mm, n = 5,897) and jumbo-head (40 or 44 mm, n = 516) groups. The median follow-up was 2.4 years (range, 0.1 to 9.8; interquartile range [IQR], 1.3 to 4.1). The primary outcome was prosthetic hip dislocation at the last clinical follow-up. Time-to-event analyses were performed using multivariable Cox models with surgeon-level random effects, adjusting for patient-specific factors, year of surgery, surgical approach, and robotic assistance.

RESULTS: In the full follow-up period, jumbo heads were not associated with a lower adjusted dislocation risk (hazard ratio (HR): 0.55 [95% confidence interval (CI): 0.13 to 2.34], P = 0.139) when compared with 36 mm heads. When analysis was truncated [...]

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Dogma Under the Knife: Challenging the Status Quo in Primary Total Knee Arthroplasty.

Padgett BM, Terhune EB, Chalmers BP, Pagnano MW, Hannon CP

INTRODUCTION: Primary total knee arthroplasty (TKA) is among the most successful procedures in orthopaedic surgery, reliably providing pain relief, functional improvement, and durable implant survivorship. Despite these achievements, approximately 10 to 20% of patients report persistent dissatisfaction following primary TKA, even in the absence of technical failure. This disconnect between implant longevity and patient-perceived success has prompted renewed examination of the foundational principles that have historically guided TKA practice. Advances in implant design, biomaterials, surgical technology, and a changing patient population have challenged long-standing dogma related to alignment philosophy, fixation strategy, bearing surface selection, and patellar resurfacing.

METHODS: This symposium manuscript reviews contemporary evidence addressing the evolving paradigms in primary TKA, with a focus on individualized alignment strategies, the growing role of cementless fixation, expanded bearing design options, and selective approaches to patellar resurfacing.

RESULTS: In each of these areas, the goal is not to promote a single new standard, but rather to provide an evidence-based framework to support thoughtful, patient-specific decision-making.

CONCLUSION: As the field continues to evolve, moving beyond rigid dogma toward personalized reconstruction strategies may repres [...]

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Revision Total Joint Arthroplasty at the Ambulatory Surgery Center: A Single Institutional Experience.

Acuña AJ, Potluri AS, Yadav AS, Alfonsi S, Forlenza EM, Della Valle CJ

INTRODUCTION: The safety and efficacy of outpatient primary total joint arthroplasty (TJA) in appropriately selected patients have been established. As rates of revision TJA (rTJA) continue to increase, there is growing interest regarding the complications and revision incidence among procedures performed in the outpatient setting. The purpose of this study was to report outcomes of revision TJA procedures performed at an ambulatory surgery center (ASC).

METHODS: A retrospective case series review of 181 patients undergoing aseptic rTJA at a free-standing ASC between January 1, 2017, and July 1, 2024, at a single large academic referral center was performed. Patients who did not have a minimum 1-year follow-up were excluded (n = 28). Our primary outcome was the incidence of complications within 90 days of rTJA, with our secondary outcome as the incidence of re-revision at final follow-up. A majority of patients underwent modular component exchanges (75.7%), with instability being the most common revision etiology (71.8%). The mean follow-up was 3.4 years (range, 1.1 to 8.6). A Kaplan-Meier survival analysis was performed to identify survivorship free of all-cause re-revision at two and five years postoperatively.

RESULTS: There was one patient (0.6%) who developed a DVT within 90 days postoperatively. There were no other medical complications seen among included patients. The [...]

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The "Distal Femoral Replacement Downgrade": Technique of Using Bicondylar Femoral Cones with a Hinge Total Knee Arthroplasty Revision in Patients Who Have Massive Distal Femoral Bone Loss as an Alternative.

Chalmers BP, Afzal S, Bhatti P, Anatone AJ, Blevins JL, Debbi EM, Sculco PK, Gausden EB

BACKGROUND: Massive femoral bone loss with collateral insufficiency in revision total knee arthroplasty (TKA) is often managed with distal femoral replacement (DFR), but this procedure sacrifices bone stock and carries notable risks. We described outcomes of a bone-preserving alternative (the "DFR downgrade") that pairs bicondylar, metadiaphyseal-engaging femoral cones with a hinge construct.

METHODS: We performed a retrospective review of consecutive patients who underwent revision TKA using a "DFR downgrade" technique for severe femoral bone loss between February 2016 and July 2024. There were twenty patients (mean age, 69 years (range, 58 to 86); 65% men) included, with a mean follow-up of 14.5 months (range, 0.9 to 43.7). Indications were aseptic loosening in 75%, infection in 10%, ligamentous instability in 10%, and arthrofibrosis in 5%. Demographics, Anderson Orthopaedic Research Institute (AORI) classification, operative details, postoperative complications, and clinical and radiographic outcomes were collected. The Kaplan-Meier method was used to estimate reoperation-free and re-revision-free survivorship. Femoral bone loss was classified as AORI IIA in 10%, IIB in 30%, and III in 60% of cases.

RESULTS: Survivorship free from revision for aseptic loosening was 100% at one and two years. Survivorship free from any re-revision was 85.1% at one and two years. Survivorshi [...]

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Outcomes of Total Hip Arthroplasty Based on Surgeon Fellowship and Case Volume.

Katanbaf R, Misch M, Aflatooni JO, Smitterberg CW, Griffin D, Nace J, Delanois R, Mont MA

BACKGROUND: Total hip arthroplasty (THA) is a highly successful procedure performed by orthopaedic surgeons of various fellowship training, case volumes, and practice types. This study aimed to compare postoperative outcomes between fellowship-trained (FT) arthroplasty and non-arthroplasty surgeons, stratified by low, intermediate, and high case volume. Outcomes were collected for up to two years postoperatively: 1) mechanical complications (dislocation, aseptic loosening, and periprosthetic fracture (PPFX)), 2) infectious complications (periprosthetic joint infection (PJI) and surgical site infection (SSI)), 3) aseptic revision, and 4) venous thromboembolism (VTE).

METHODS: We retrospectively queried a national all-payer database to identify 457,372 patients who had primary THA between 2010 and 2021. Using National Provider Identifiers, the number of surgeons was categorized by fellowship training (arthroplasty versus non-arthroplasty) and stratified by total case volume during the study period (high, intermediate, and low). Propensity score matching was performed with the respective FT group as the basis of comparison.

RESULTS: High-volume FT surgeons demonstrated lower hip dislocation rates compared with all NFT cohorts at 90 days (0.8 versus 0.9 to 1.2%), one year (1.1 versus 1.2 to 1.5%), and two years (1.2 versus 1.3 to 1.7%) (all $P < 0.001$). High-volume FT surgeons als [...]

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Functional Outcomes, Complications, and Survivorship After Simultaneous Bilateral Total Hip Arthroplasty: A Retrospective Cohort Study with a Median Follow-up of 4.6 Years.

Diaz Dilernia F, Vasarhelyi E, Lanting B, Naudie D, McCalden R, MacDonald S, Howard J

BACKGROUND: Simultaneous bilateral total hip arthroplasty (sbTHA) offers the advantages of single-stage treatment for end-stage bilateral hip disease, but raises concerns about increased perioperative risk and implant survivorship. This study aimed to evaluate functional outcomes, complications, and implant survivorship following sbTHA under contemporary perioperative protocols with a median follow-up of 4.6 years.

METHODS: We conducted a retrospective study of 92 patients (184 hips) who underwent elective sbTHA at a single institution between 2008 and 2019. Their median age was 60 years (interquartile range [IQR], 52 to 68). There were 48 men (52.2%), and the median follow-up was 4.6 years (IQR, 3.2 to 7.0). The primary outcomes included functional scores, complications, readmissions, and implant survivorship.

RESULTS: There was a significant improvement in functional scores and pain evaluation at final follow-up ($P < 0.001$). The overall complication rate was 6.5% (12 of 184 hips). Transfusion was required in 6.5% of patients (six of 92). When revision surgery was used as the endpoint, the implant survivorship was 95.7%.

CONCLUSIONS: The sbTHA provides excellent functional outcomes and survivorship at a median follow-up of 4.6 years, with an acceptable complication profile in carefully selected patients managed with modern perioperative and blood management protocols.

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How Do Operating Room Timings Differ Between Anterior and Posterior Approaches for Total Hip Arthroplasty.

Ruff G, Di Pauli von Treuheim T, Sarfraz A, Metko K, Rozell JC, Schwarzkopf R, Aggarwal VK

BACKGROUND: Total hip arthroplasty (THA) is commonly performed using either an anterior- or posterior-based approach (AA or PA), with similar long-term outcomes. However, operative timings, including time for patient positioning, device implantation, wound closure, and total operating room (OR) time, may differ between these approaches. This study assessed differences in various OR timings between AA and PA for primary THA.

METHODS: A retrospective review was performed of all patients who underwent primary, unilateral, elective THA at our large urban academic medical center between January 1, 2019, and January 31, 2025. Patient demographics and operative parameters were determined from clinical and operative records. Surgeons who began practicing during the study period had their first 50 cases excluded. Perioperative timings were compared between approaches. Multivariable regression analyses controlled for age, American Society of Anesthesiologists status, body mass index, cementation, robotic use, navigation assistance use, discharge disposition, sex, smoking status, and year of surgery. There were 8,120 procedures included (4,670 PA and 3,450 AA).

RESULTS: The AA had shorter total OR time (156.7 versus 167.4 minutes, $P < 0.001$). Regression analysis found AA was associated with longer implantation time (+3.3 minutes, $P < 0.001$), but shorter set-up time (-2.0 minutes, $P < 0$. [...]

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Factor XI and XIa Inhibitors for Venous Thromboembolism Prophylaxis After Total Knee Arthroplasty: A Review From Mechanism to Phase III Trials.

Smitterberg CW, Ando W, Chen AF, Garvin KL, Hamilton WG, Lee GC, Lieberman JR, Lombardi AV et al.

Venous thromboembolism (VTE) prophylaxis after total knee arthroplasty (TKA) has improved substantially, yet the field continues to balance two competing priorities: reduction of symptomatic deep vein thrombosis and pulmonary embolism while minimizing bleeding, wound complications, and downstream infection risk. Contemporary practice has shifted toward aspirin as prophylaxis for many low-risk patients after TKA, with direct oral anticoagulants (DOAC) and low-molecular-weight heparin (LMWH) reserved for higher-risk patients; however, these agents still affect common-pathway thrombin generation and can increase bleeding or compromise wound healing. Aspirin is not an anticoagulant and, despite its widespread use, does not directly address coagulation-driven thrombosis while still increasing bleeding risk, particularly gastrointestinal bleeding. Factor XI (FXI) and activated FXI (FXIa) sit upstream in the intrinsic pathway and may contribute disproportionately to thrombosis relative to hemostasis. In orthopaedic surgery, multiple Phase II programs, including antisense oligonucleotide knockdown of FXI, monoclonal antibody blockade of FXI or FXIa, and small-molecule FXIa inhibitors, have demonstrated large reductions in venographic VTE with low bleeding profiles comparable to enoxaparin. These data have accelerated late-stage development, as there are two ongoing pivotal Phase III pr [...]

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Intraosseous Sustained-Release Anesthetics Promote Early Ambulation and Functional Recovery after Total Knee Arthroplasty.

Huang J, Abdullah Ezzi SH, Wu S, Ali Alshabi MM, Zhang W, Adil MM, Cao X

BACKGROUND: Postoperative pain is a primary impediment to functional recovery in patients who undergo total knee arthroplasty (TKA). This work aimed to evaluate the efficacy of intraosseous sustained-release anesthetics, administered into the subchondral trabecular space before the insertion of a prosthesis during TKA, in promoting

early ambulation and functional recovery.

METHODS: This randomized controlled open-label study was carried out on 110 patients who had a mean age of 75 years (range, 65 to 90). Patients were subdivided into two groups: Local infiltration anesthesia (LIA) group : 10 mL of sustained-release anesthetic injection solution was taken, and infiltration anesthesia was applied in the joint and around the incision (two mL was injected into the posterior joint capsule and lateral collateral ligaments, two mL was injected into the knee-extension device and the parapatellar, and six mL was injected into the periarticular soft tissues); and Local infiltration anesthesia + intraosseous infiltration anesthesia (TIA) or Group LIA plus TIA: a sustained-release anesthetic injection solution of 20 mL, 10 mL as described before plus intraosseous infiltration anesthesia on the osteotomy surface of the tibia (10 mL).

RESULTS: Visual analog scale (VAS), at rest and during walking, was significantly lower at days one and three in group LIA plus TIA than in group LIA ($P < 0$. [...])

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Comparing Clinical and Radiological Outcomes for a Three-Dimensional Printed Highly-Porous Cup and a Hydroxyapatite-Coated Non-Highly-Porous Cup in Total Hip Arthroplasty Using Propensity Score Matching.

Okazaki T, Imagama T, Matsuki Y, Kaneoka T, Kawakami T, Sakai T

BACKGROUND: The use of highly-porous acetabular components in total hip arthroplasty (THA) has gained popularity. To elucidate the clinical and radiological outcomes of three-dimensional (3D)-printing highly porous cups, we compared highly-porous cups and hydroxyapatite (HA)-coated non-highly-porous cups in computed tomography (CT)-based navigated THA.

METHODS: We retrospectively investigated 196 hips of 175 patients (the highly-porous group, 112 hips of 97 patients; the HA-porous group, 84 hips of 78 patients) who underwent THA using a CT-based navigation system. After propensity score matching for age, sex, diagnosis, and surgical approach, each group comprised 66 hips. Clinical outcomes were evaluated using the Japanese Orthopaedic Association hip (JOA) score and Japanese Orthopaedic Association Hip Disease Evaluation Questionnaire (JHEQ) score preoperatively and at six, 12, and 24 months postoperatively. Radiological outcomes for cup orientation, positioning, initial gap, migration, loosening, and radiolucent line (RLLs) were assessed. Cup placement was measured using 3D templating software.

RESULTS: Clinical outcomes did not differ significantly between groups after two years. The mean absolute error of cup orientation did not differ significantly between the two groups (inclination: $2.9 \pm 2.3^\circ$ versus $2.6 \pm 2.2^\circ$, respectively; anteversion: $2.7 \pm 2.0^\circ$ versus $2.7 \pm 2.0^\circ$, r [...])

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Aspirin is Associated with Improved Venous Thromboembolism Prophylaxis Compared with Anticoagulants After Total Hip Arthroplasty in Patients Who Have Prior Hematologic Malignancy.

Paul BR, Verhey JT, Tarabichi S, Van Schuyver PR, Christopher ZK, Bingham JS, Spangehl MJ

BACKGROUND: Patients who have prior hematologic malignancies undergoing total hip arthroplasty (THA) have elevated venous thromboembolism (VTE) risk. Although guidelines favor low-molecular-weight heparin (LMWH) or direct oral anticoagulants (DOACs) in high-risk patients, these agents may increase bleeding and wound complications. Evidence supporting aspirin in this population remains limited.

METHODS: A national administrative database identified patients diagnosed with a hematologic malignancy and underwent THA between 2010 and 2023. Patients were stratified by postoperative VTE prophylaxis. After propensity score matching, 1,043 patients who received aspirin were matched to 1,484 patients who received non-aspirin anticoagulation on demographics and comorbidities. The primary outcome was 90-day VTE. The secondary outcomes included postoperative complications, readmissions, and mortalities. Subgroup analyses evaluated malignancy-to-THA timing and anticoagulant classes.

RESULTS: At 90 days, aspirin was associated with lower DVT rates compared with non-aspirin prophylaxis (0.3 versus 1.1%; odds ratio (OR) 0.25 [0.07 to 0.85]; $P = 0.03$). There were no differences in pulmonary embolism or mortality. Aspirin was associated with fewer emergency department visits (OR 0.62 [0.48 to 0.80]; $P < 0.001$), readmissions (OR 0.57 [0.41 to 0.80]; $P = 0.001$), transfusions (OR 0.44 [0.25 to 0. [...])



Does the surgical sequence of lumbar spine fusion and total hip arthroplasty affect the dislocation and revision risk? A meta-analysis and systematic review.

Liu F, Jia C, Wu X, Zhou Y

BACKGROUND: Patients who underwent primary total hip arthroplasty (THA) after lumbar spine fusion (LSF) had a higher incidence of complications compared with those who had not previously undergone LSF. This study aimed to determine whether the timing of THA before or after LSF impacts the incidence of THA complications.

METHODS: Following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, we searched PubMed, Web of science, Embase, and Cochrane Library from their inception until 1 October 2025. Data from all eligible published studies meeting the inclusion criteria were extracted and subsequently analyzed using a random-effects model to calculate the rates of all-cause revision, dislocation, periprosthetic joint infections, and aseptic loosening.

RESULTS: Our analysis included 10 studies involving 27,605 patients who underwent THA followed by LSF and 32,002 patients who received LSF prior to THA. There are no significant differences in the rates of dislocation (odds ratio [OR] 1.19, 95% confidence interval (CI) 0.73-1.86, $p = 0.45$), periprosthetic joint infections (OR 0.91, 95% CI 0.52-1.62, $p = 0.76$), and aseptic loosening (OR 0.89, 95% CI 0.74-1.08, $p = 0.23$) between the two surgical sequences, while for all-cause revision, the OR was (OR 1.01, 95% CI 0.70-1.47, $p = 0.94$), which reached statistical significance.

CONCLUSIONS: Compare [...]

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Letter to the Editor Regarding "Proton Pump Inhibitors Are Associated With Increased Risk of Site-Specific Nonunion After Open Reduction Internal Fixation".

Ayas O, Acar A

[PubMed](#) · [DOI](#)

Response to "Competing-Risks Re-estimation of Five-Year Second Hip Fracture Risk: Accounting for Death as a Competing Event".

Konda SR

[PubMed](#) · [DOI](#)

Response Letter to the Editor Regarding Article "Orthopaedic Trauma Association Annual Meetings Consistently Provide Academic Value for Attendees".

Greenhill DA, Lundy D, Imam N, Jain N

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Letter to the Editor Regarding Article "Orthopaedic Trauma Association Annual Meetings Consistently Provide Academic Value for Attendees".

Blomberg JR

A Primer on Statistical Certainty and its Distinction from Clinical Importance.

Ebrahimzadeh E, Slobogean GP, Ricci WM, McKellop H



Archives of Orthopaedic & Trauma Surgery

Volume 146, Issue 1 · June 2026 · 3 new articles

Early-onset rice body synovitis of the knee in children under three years: a case series and review of the literature.

Zhu L, Liu C, Liu Y, Mao B, Huang Y, Wen J, Liu Y, Qian T

OBJECTIVE: Rice body synovitis reports have been rarely reported in children under three years of age. We aimed to explore the clinical characteristics and treatment experience of rice body-like synovitis in children's knees.

METHODS: We retrospectively analyzed the clinical and imaging data of 4 cases (5 knees) of rice body-like synovitis in children treated at our hospital from January 2014 to December 2021. Among the patients, 1 was male and 3 were female, with an average age of 22.5 months. The average length of symptoms before admission was 1.86 months. Clinical presentations included swelling and limping, with no fever or rash.

RESULTS: Three patients with unilateral onset and one patient with bilateral onset underwent unilateral arthroscopic excision of synovial lesions. All 4 patients had pathological findings consistent with chronic inflammation, leading to a postoperative diagnosis of juvenile idiopathic arthritis (JIA). Postoperatively, all patients received standardized pharmacological treatment and were followed for 24 to 48 months in collaboration with the rheumatology department. None of the 4 knees undergoing surgery experienced recurrence, and the symptoms of the knee treated conservatively resolved spontaneously. In the last follow-up session, MRI and ultrasound showed only mild synovial thickening, with effusion and loose bodies disappearing.

CONCLUSION: R [...]

Extravertebral temperature exposure during spinal radiofrequency ablation: an experimental surrogate assessment of neural injury risk.

Weigert A, Hoffmann RT, Büttner A, Zaccaria R, Wegener V, Holzapfel BM, Wegener B

BACKGROUND: Radiofrequency ablation (RFA) has become an important minimally invasive option for the treatment of painful spinal tumors and benign bone lesions. While its effectiveness within the vertebral body is well established, there is ongoing concern about unintended heat exposure of nearby neural structures. Experimental data describing how heat propagates beyond the vertebra during spinal RFA are still limited. This experimental in vitro study aims to assess the extravertebral temperature exposure during spinal radiofrequency ablation as a surrogate parameter for potential neural injury risk under controlled conditions.

METHODS: This experimental in vitro study was performed on four fresh-frozen human lumbar spines. Radiofrequency ablation was carried out using a commercially available system and standard clinical protocols. Two anatomical configurations were examined: ablation within the vertebral body and ablation within the pedicle. Temperatures were measured at predefined locations outside the vertebra, including the spinal canal and cortical boundaries, using multiple thermocouples. To aid interpretation of radiofrequency-related measurement artifacts, an additional control setup using an electrothermal heating system without radiofrequency energy was included. Temperature data were evaluated descriptively.

RESULTS: In the control experiments, temperature profiles [...]

Giant cell tumor of bone: exploring HtrA1 as a potential predictor of recurrence.

INTRODUCTION: Giant cell tumor of bone (GCTB) is a locally aggressive tumor with a considerable risk of recurrence, but reliable predictive markers are limited. High-temperature requirement A serine peptidase 1 (HtrA1) has been implicated in the progression of several cancers and may also play a role in GCTB recurrence.

MATERIALS AND METHODS: This retrospective study included 34 patients with GCTB treated with curettage between 2002 and 2016. HtrA1 expression was evaluated separately in giant cells (GC) and mononuclear cells (MNC) using immunohistochemistry. Based on high or low expression in each cell type, patients were categorized into four expression patterns: High GC / High MNC, High GC / Low MNC, Low GC / High MNC, and Low GC / Low MNC. Clinical data, including recurrence status and time to recurrence, were analyzed.

RESULTS: The High GC / Low MNC group showed the highest recurrence rate at 71.4%, compared with 28.6% in the High GC / High MNC group and 31.6% in the Low GC / Low MNC group. No recurrence was observed in the Low GC / High MNC group, which included only one patient. The High GC / Low MNC group showed a non-significant trend toward higher recurrence compared with all other groups combined, with a similar trend toward shorter recurrence-free survival.

CONCLUSION: HtrA1 expression patterns in GCTB may provide preliminary insight into recurrence risk. Although [...]

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Injury

Volume 57, Issue 8 · June 2026 · 15 new articles

The forequarter lateral implosion injury: A diagnostic paradigm and Ch-Sh (ch-sh) classification.

George A, Paull TZ, Winter JD, Levine J, Cole PA

BACKGROUND: The lateral impaction force against the shoulder often causes multiple lesions to the forequarter including the chest wall. The purpose of this study is to describe the injury patterns and associated pathology in patients with combined chest wall and shoulder girdle injuries- forequarter lateral implosion injury and organize the findings into a classification framework to predict morbidity.

METHODS: Three hundred fifty-four consecutive patients with two or more rib fractures and ipsilateral shoulder girdle injury from a Level 1 trauma center, between 2018-2022, were retrospectively reviewed. Ch-Sh is pronounced ch sh. Chest injuries were grouped into three categories: Ch1: 2-4 fractured ribs, Ch2: 5 + fractured ribs, or Ch3: 2 + fractured ribs with contralateral rib fractures and/or sternal fracture. Shoulder girdle injuries were grouped into: Sh1: clavicle or acromion fracture, and/or AC/SC joint dislocation, Sh2: scapula and/or proximal humerus fracture, or Sh3: combination of Sh1 and Sh2 injuries (clavicular chain and scapula/humerus). There were 9 total subgroups. Demographics, injury characteristics, associated injuries, and outcomes were recorded. Inter and intraobserver reliability of the classification system was studied.

RESULTS: As chest wall injuries scaled from Ch1 to Ch3, there were significantly higher rates of ICU admission, pneumothorax, pulmonar [...]

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The fate of stage I Masquelet cement spacers in long bone reconstruction for acute fractures.

Dekle JK, Sisk CK, Litten RM, Alcaide DM, McIlwain RN, Wilson AL, Manson HC, Spitler CA et al.

INTRODUCTION: The Masquelet technique is widely used for reconstruction of segmental long bone defects following acute fractures; however, not all patients progress to the second-stage bone grafting procedure. The purpose of this study was to characterize Stage II completion following Stage I Masquelet cement spacer placement for acute long bone fractures, compare characteristics and outcomes between patients who underwent Stage I only versus completed Stage II reconstruction, and identify factors associated with progression to Stage II bone grafting.

METHODS: A retrospective review was conducted of adult patients who underwent Masquelet reconstruction for acute long bone fractures at a Level I trauma center between 2014 and 2024. Patients were stratified by completion of Stage II reconstruction. Demographic, injury-related, surgical, and clinical outcome variables were collected. Multivariable logistic regression was performed to identify factors independently associated with Stage II completion.

RESULTS: A total of 156 patients met inclusion criteria, of which 48 (30.8%) underwent Stage I only and 108 (69.2%) completed both stages. Patients who remained at Stage I underwent fewer orthopaedic operations (3.2 vs. 4.2, $p = 0.001$) and initial bony defect size did not differ between groups (84.1 vs. 78.8 mm, $p = 0.423$). Among patients with at least 6 months of follow-up, rates o [...]

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Rethinking the Lisfranc injury: Rationale for and development of a new classification system for Lisfranc injuries.

Cawley WCF, Hwang P, Mittal R, Zochowski Y, Darwish I, O'Keefe R, Grogan J, Sivakumar B et al.

BACKGROUND: The Lisfranc joint comprises the articulations between the cuneiforms and bases of the metatarsals and is susceptible to a variety of complex injuries which are frequently missed or diagnosed late. This can result in morbidity with poor outcomes, and renders it challenging to standardise management and allow prognostication. Existing classification systems, including the Myerson-Hardcastle (MHC) and Nunley & Vertullo classifications, do not consider the propagation of force through the midfoot, which is key to identifying structures which may require stabilization. They also demonstrate poor clinical utility due to poor correlation with outcomes. This study aimed to develop a new classification system for Lisfranc injuries using data from a large single-surgeon case series from a tertiary level trauma hospital in Australia. The primary outcome measures for this study were inter- and intra-observer reliability.

METHODS: The Suthersan classification was developed to address shortfalls in current classification systems. Patient records were manually searched for Lisfranc injuries. Direct 'crush' Lisfranc injuries were excluded due to unpredictable injury patterns. Indirect Lisfranc injuries were included in analysis. Two orthopaedics foot and ankle surgeons and one hand surgeon reviewed patient radiographs and classified them using the MHC and Suthersan classification [...]

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Injury and trauma care training in global health: A scoping review of costs and cost-effectiveness across resource settings.

Phan HT, Nguyen BT, Zhao T, Phung LT, Keakabetse TR, Dinh TTN, Nelson MR, Skinner M et al.

INTRODUCTION: Injury is a leading cause of death and disability worldwide, with a disproportionate burden in low- and middle-income countries, where health resources are constrained and funding decisions are highly cost-sensitive. Injury and trauma care training programs have demonstrated their effectiveness in enhancing the capacity of care providers to treat injured patients, but evidence on their costs and cost-effectiveness remains limited and fragmented. This scoping review aims to map and summarise current literature on costs and cost-effectiveness of injury and trauma care training across different resource settings.

METHODS: We conducted a systematic search on PubMed, MEDLINE Complete, Embase via Ovid, Web of Science, EconLit and Emcare. Papers published in English that reported implementation costs or cost-effectiveness of trauma care training programs were included. Cost data were converted and presented as USD 2024.

RESULTS: Twenty-eight studies were included, predominantly in low- and middle-income countries. Implementation costs varied considerably, starting from USD 3.70 (Lay-First-Responder) to USD 1433.5 (Advanced Trauma Life Support). Foreign instructor involvement was a particularly significant cost driver, accounting for 38-85% of total implementation costs in some programmes. Training-of-trainers and cascading models were effective in reducing costs, with [...]

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Do reduced screws have a significant impact on fixation in medial open wedge high tibial osteotomy? A comparative fatigue and stability analysis.

Özkaya M

INTRODUCTION: Medial opening wedge high tibial osteotomy generally employs T-shaped plates with three or more proximal screws to ensure stabilization. This study aims to evaluate the biomechanical necessity of the standard three screw proximal configuration of the T-shaped plates and investigates whether fixations with a reduced number of proximal screws provide sufficient stability and fatigue life under diverse loading scenarios using finite element analysis.

MATERIALS AND METHODS: Three fixation models were analyzed, a standard T-shaped plate with three proximal screws in the first row and two modified models with two proximal screws in the first row with or without an empty hole. Models were subjected to physiological axial loading, centrally, anteriorly, or posteriorly applied in the sagittal plane, to measure total deformation, displacement of the tibial plateau, fixation stiffness, and von-Mises stress. Furthermore, a fatigue analysis was conducted to evaluate the endurance of fixations under these varying loading conditions.

RESULTS: The reduction in proximal screws resulted in a marginal stiffness decrease of less than 2%, with values remaining above 7700 N/mm under centric loading. Total deformation and tibial plateau displacement showed negligible differences between two screw and three screw configurations. In addition, all peak stresses remained well below the [...]

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Moving beyond mortality measurements after road traffic injuries: Ten-year death risk after road traffic injuries.

Cherchiglia ML, Mougin V, Gray A, Wool EE, Malta DC, Melo APS, de Assis Acúrcio F, Junior AAG et al.

Road traffic injuries (RTIs) represent a significant global public health issue, particularly in low- and middle-income countries like Brazil. Despite the high mortality rates and associated costs, there is limited comprehensive literature addressing long-term mortality resulting from RTIs. Most research tends to focus on rates and hospitalization figures while neglecting the connection between hospitalizations and causes of death, partly due to the challenges of follow-up studies. This study employed record-linkage methodology to create a database that includes both hospitalization and mortality records. The aim was to estimate all-cause and cause-specific relative risks (RR) of death among inpatients due to road traffic accidents, compared to those hospitalized for other reasons, over a time span of three months to ten years. A retrospective cohort study examined hospitalization data for RTIs and mortality from 2000 to 2015 using Brazil's Unified Health System (Sistema Único de Saúde - SUS in Portuguese), excluding any obstetrics-related admissions and deaths, and calculated RR stratified by sex and age over 15 years old. The findings indicated a significantly higher mortality risk of 3.23 (95% Uncertainty Interval: 3.08- 3.39) for individuals previously hospitalized for RTIs compared to others during follow-up. Males aged 15- 29 and 30- 59, especially those with motorcycle, [...]

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MAGPIE study author's reply.

Hartshorn S, Barrett MJ, Lyttle MD, Yee SA

[PubMed](#) · [DOI](#)

Can we conclude "Psychosocial intervention is not required"? Methodological reflections regarding a randomized controlled trial on post-traumatic limb amputation.

Zhao FY, Zhang WJ, Fu QQ, Zhu JY, Qian J

[PubMed](#) · [DOI](#)

Overutilization of helicopter emergency medical services compared to ground transport in moderate-severe penetrating abdominal trauma & associated outcomes.

Hus A, Brown A, Bundschu I, Prashar S, Kumar S, Indorewala Y, Elkbuli A

BACKGROUND: Helicopter emergency medical services (HEMS) play a crucial role in trauma transport, accounting for approximately 550,000 transportations annually. However, HEMS holds a disproportionate share of trauma causes, highlighting its potential overutilization.

METHODS: This retrospective cohort study utilized the ACS-TQIP-PUF dataset (2017-2020) to evaluate outcomes in adult (≥ 18 years) trauma patients in hemorrhagic shock with moderate-to-severe (AIS abdomen ≥ 2 , ISS ≥ 15) penetrating abdominal injuries who activated massive transfusion protocol and were transported from scene to the hospital via either HEMS or GEMS. Patients were stratified by scene time and trauma center level. Primary outcomes included mortality, and secondary outcomes included transfusion requirements, intervention rate, and complications.

RESULTS: Patients transported via Ground Emergency Medical Services (GEMS) compared to HEMS were associated with significantly higher odds of 24-hour mortality (2.028, 95% CI: 1.185-3.472, $p = 0.010$, SE: 0.274). However, no significant association with in-hospital mortality (1.355, 95% CI: 0.901-2.038, $p = 0.144$, SE: 0.208) was observed. Patients with a scene time of < 10 min demonstrated comparable outcomes, with no significant association with 24-hour or in-hospital mortality.

CONCLUSION: This national analysis demonstrates that outcomes of adult patients in h [...]

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Early postoperative anxiety is associated with patient-reported recovery after hip fracture: A PROMIS-based longitudinal cohort study.

Achebe C, Cureton R, Rothberg D, Higgins T, Marchand L, Haller J

BACKGROUND: Hip fractures in older adults result in substantial functional decline. While physical factors influencing recovery are well established, the role of early postoperative psychological distress remains incompletely understood. We examined whether early postoperative anxiety was associated with achievement of minimal clinically important difference (MCID) in patient-reported recovery after hip fracture.

METHODS: We conducted a retrospective cohort study of patients aged > 55 years with native hip fracture treated operatively between 2015 and 2024 who completed PROMIS assessments. Baseline PROMIS assessments were defined as the first available domain-specific assessment during postoperative days 0-30. The primary outcome was MCID achievement in PROMIS Physical Function (PF) and Pain Interference (PI) using the final available assessment from postoperative days 31-365. A late-recovery sensitivity analysis restricted final follow-up to postoperative days 91-365. MCID was defined as a $> =5$ -point improvement in PF or $> =5$ -point reduction in PI. Multivariable logistic regression adjusted for baseline domain score, baseline anxiety, age, and Charlson Comorbidity Index (CCI).

RESULTS: The primary PF cohort included 304 patients, of whom 171 (56.3%) achieved MCID. The primary PI cohort included 285 patients, of whom 169 (59.3%) achieved MCID. Higher early postoperative anxie [...]

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Association between neighborhood opportunity and incidence of pediatric trauma.

Monroy A, Tan JM, Arbogast H, Spurrier R, Nager A, Chaudhari PP

OBJECTIVE: To describe the epidemiology of pediatric trauma and its association with neighborhood opportunity in a large, urban metropolitan area.

STUDY DESIGN: This was a multicenter, retrospective, cross-sectional study of pediatric trauma activations in Los Angeles (LA) County between 2010 and 2021. Children under 18 years old with a home zip code in LA County were eligible. Our main exposure was the categorical Child Opportunity Index 2.0 (COI) score for a child's home zip code. The primary outcome was trauma activations per 100,000 children. Our secondary outcomes were variation in injury mechanism, severity and disposition by COI.

RESULTS: The final sample was comprised of 28,034 patients. The median age was 13.0 years (IQR 7.0, 16.0). Mortality was 1.8% ($n = 493$). Overall, geographic areas with lower COI overlapped with areas of higher trauma incidence. Falls and sports-related injuries were more common in higher COI areas (all $p < 0.001$). Conversely, being a pedestrian or bicyclist struck, firearm injuries and assaults/stabbings were more common in lower COI areas (all $p < 0.001$).

CONCLUSION: The incidence of pediatric trauma varies based on the COI of a child's home zip code. Our findings can help inform resource allocation for injury prevention programs targeted to specific neighborhoods at greatest risk.

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Mechanism of injury not associated with acute compartment syndrome in Tibia fractures.

Egbe S, Marks D, Christiano AV, Erdman MK, Wolf JM, Strelzow J

OBJECTIVES: To report the incidence of Acute Compartment Syndrome (ACS) after low-energy, ballistic tibial fractures and examine associations between injury mechanism, fracture location and morphology, and patient characteristics.

METHODS: Design: Retrospective Review Setting: Single-center, large, urban, level 1 trauma center Patient Selection Criteria: Consecutive skeletally mature patients presenting with a tibial fracture [OTA-41, OTA-42, OTA-43] were evaluated.

OUTCOME MEASURES AND COMPARISONS: Patients were stratified into blunt and ballistic fracture cohorts based on mechanism of injury. Patient sociodemographics, diagnosis of ACS, and established risk factors for the development of ACS (age, gender, fracture location, and morphology) were collected.

RESULTS: Over the study period, 281 patients were included. Study participants had a mean age of 37 years (range, 15-98) with the majority being male, (70%). Based on injury mechanism, 67% (n = 187) were blunt injuries, and 32% (n = 94) were ballistic injuries. Fractures were categorized according to the AO/OTA classification and included 69 [OTA-41], 163 [OTA-42], and 47 [OTA-43] fractures [1]. The overall incidence of ACS in this population was 2.1% (n = 6/281), with a incidence of 1% (n = 1/94) in the ballistic cohort and 2.7% (n = 5/187) in the blunt cohort. Mechanism of injury, sex, age, and open fractures did not sh [...]

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Psychological outlook and recovery after orthopaedic trauma: Evidence and implications for Central India and other low-resource settings.

Salphale YS, Saodekar H, Kimmatkar N, Shinde G, Kothari P, Salphale AY

BACKGROUND: Orthopaedic trauma is frequently accompanied by psychological distress, including depression, anxiety, and post-traumatic stress, particularly in low- and middle-income countries. Beyond injury severity and surgical factors, psychological outlook-encompassing negative cognitions (e.g., catastrophising) and positive constructs (e.g., optimism, resilience, social support)-may substantially influence pain, disability, and functional recovery [3,4,6,17,18].

OBJECTIVES: To narratively review evidence on the impact of psychological outlook on clinical, functional, and psychosocial outcomes after orthopaedic trauma, with emphasis on India, Central India, and comparable low-resource settings [4,10,17].

METHODS: A structured narrative review of English-language studies (January 2000-October 2025) was conducted using PubMed, Scopus, and Google Scholar. Observational and interventional studies examining associations between psychological factors and pain, disability, function, quality of life, or rehabilitation outcomes in adult orthopaedic trauma populations were included [4,6,15].

RESULTS: Across diverse settings, approximately one-quarter to one-half of orthopaedic trauma patients experience clinically significant psychological distress, which consistently predicts higher pain, greater disability, and poorer quality of life independent of injury characteristics [2-4,14,1 [...]

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Satisfaction with nursing care among Jordanian burn patients admitted to burn critical care units: A cross-sectional study.

Al-Ghabeesh S, Alnaeem M, Qaddoura N, Ali BA, Al-Masalha HA

BACKGROUND: Patient satisfaction with nursing care is a key indicator of healthcare quality in burn intensive care units (ICUs). However, limited evidence exists regarding predictors of satisfaction among burn patients in Jordan.

OBJECTIVE: The aim of this study was to assess satisfaction with nursing care among burn patients admitted to a burn intensive care unit in Jordan and to identify demographic and clinical predictors associated with satisfaction.

METHODS: A descriptive cross-sectional study was conducted between October 2024 and May 2025 in a specialized Burn ICU at a major military hospital in Jordan. Adult burn patients (n = 146) admitted for ≥ 48 h were recruited using convenience sampling. Satisfaction was measured using the Arabic version of the Patient Satisfaction with Nursing Care Quality Questionnaire (PSNCQQ). Independent t-tests, ANOVA, Pearson correlation, and multiple linear regression were used. Significance was set at $p < 0.05$.

RESULTS: The mean satisfaction score was 48.88 (SD = 13.83), indicating moderate satisfaction. Higher income was associated with greater satisfaction (B = -6.529, 95% CI: -11.65 to -1.40, $p = 0.014$). Facial deformity predicted lower satisfaction (B = 7.743, 95% CI: 1.30-14.18, $p = 0.019$). Age showed a weak but significant correlation ($r = -0.173$, $p = 0.037$).

CONCLUSION: Satisfaction among burn ICU patients was moderate and infl [...]

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Flexible intramedullary nailing in the treatment of paediatric forearm fractures: 15-years' experience at a level one trauma centre, comparison of implant retention versus removal and cost-effectiveness analysis.

Ismayl G, Alabbas A, Howard A, Fishlock A, Sabouni MY

INTRODUCTION: Flexible intramedullary nailing (FIN) is a common treatment method for paediatric forearm fractures. They are routinely removed after fracture healing. This study aims to assess the 15-years' experience of managing paediatric forearm fractures with FIN, with comparison between the outcomes of implant retention versus removal and cost analysis. This is the first study on FIN retention in paediatric forearm fractures.

METHODS: This was retrospective cohort study for paediatric forearm fractures treated with FIN between 2009 and 2024 at Leeds General Infirmary, a level one trauma centre in the United Kingdom. We noted a significant decline in scheduled elective removal of forearm FIN following the Coronavirus pandemic (COVID-19). The cohort was therefore divided into two comparative groups: pre-COVID (routine implant "removal") and post-COVID (routine implant "retention"). Primary outcome measured was incidence of secondary (unplanned) surgery. Secondary outcomes measured were indications for surgery, time from index procedure to FIN removal and subjective patient/parents' satisfaction.

RESULTS: A total of 268 patients were included in the study. The pre-COVID group comprised of 212 patients, and the post-COVID group 56 patients. Elective implant removal was performed in 68.9% (n = 146) of patients in the earlier group and 3.6% (n = 2) in the latter. Secondary surg [...]

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Denis pattern-dependent biomechanical behavior of posterior trans-ilial plate fixation compared with bilateral triangular osteosynthesis in unilateral sacral fractures: a finite element study.

Yang J, Böker KO, Li C, Schilling AF, Ritter Z, Acharya M, Lehmann W

OBJECTIVE: Posterior trans-ilial plate osteosynthesis (TPO) is commonly used for the stabilization of unilateral vertically unstable sacral fractures. However, its biomechanical performance across different Denis fracture patterns remains unclear. This study aimed to investigate the fracture pattern-dependent biomechanical behavior of posterior trans-ilial plate osteosynthesis (TPO) in unilateral vertically unstable sacral fractures, using bilateral triangular osteosynthesis (BTO) as a high-stability reference construct.

METHODS: A validated three-dimensional finite element model of the lumbopelvic complex was developed to simulate unilateral vertically unstable sacral fractures corresponding to Denis type I, II, and III patterns. Two posterior fixation strategies, TPO and BTO, were constructed for each fracture type. Seven physiological loading conditions were applied, including standing, flexion, extension, axial rotation, and lateral bending. Von Mises stress distribution and fracture displacement were evaluated. Fracture micromotion was quantified by calculating relative displacement changes between paired points along the fracture gap.

RESULTS: When interpreted relative to the high-stability BTO reference construct, TPO showed fracture pattern-dependent variation in fracture displacement. The difference was most pronounced in Denis type I fractures and progressively decreased [...]

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Factors associated with in-hospital mortality in necrotising soft tissue infections. a multicentre retrospective cohort study.

Podda M, Ceresoli M, Viridis F, Rosa F, Pilia T, Murzi V, Dessì A, Tedesco S et al.

PURPOSE: Necrotising soft tissue infections (NSTIs) are rare but life-threatening conditions associated with high mortality rates. This multicentre study aimed to identify admission variables associated with in-hospital mortality.

METHODS: This retrospective study included adult patients with surgically confirmed NSTIs treated at four high-volume academic referral centres in Italy between 2010 and 2024. Demographic, clinical, physiological, and laboratory variables available at hospital admission were analysed. Categorical variables were compared between survivors and non-survivors using the chi-square test or Fisher's exact test, while quantitative variables were compared using the Student's t test or Mann-Whitney U test. Multivariable logistic regression analyses were performed to identify independent predictors of in-hospital mortality. Results were reported as ORs with 95%CI. A prognostic nomogram was developed from the final multivariable model, including variables independently associated with mortality.

RESULTS: A total of 379 patients were included. In-hospital mortality was 16.7%. In subgroup comparisons, mortality was 17.0% in necrotising fasciitis and 22.4% in Fournier's gangrene ($P = 0.275$). In the overall NSTI population, age (aOR 1.061, 95%CI 1.037-1.087) and serum creatinine at admission (aOR 1.301, 95%CI 1.040-1.629) were independently associated with mortality [...]

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Influence of frailty on short-term mortality in older patients with multiple trauma in the emergency department.

Çelik ■, Toker A, Biçer S, Cebecio ■, Lu ■, K ■

PURPOSE: To investigate the association between frailty and 28-day mortality in older patients presenting to the emergency department with multiple trauma and to assess the discriminative performance of PRISMA-7 and the Trauma-Specific Frailty Index (TSFI) in frailty assessment in relation to commonly used trauma severity scores.

METHODS: This prospective observational study included patients aged ≥ 65 years with multiple trauma admitted to the emergency department between July 1 and August 31, 2024. Frailty was assessed using PRISMA-7 and TSFI. Patients were stratified into frail and non-frail groups and further categorized according to age groups (65-74, 75-84, and ≥ 85 years). Clinical characteristics, trauma severity scores, and outcomes, including 28-day mortality, were compared. Receiver operating characteristic (ROC) analysis was used to evaluate discriminative performance, and logistic regression analysis was performed to assess the association between scores and mortality.

RESULTS: A total of 144 patients were included. Frailty was significantly associated with increased 28-day mortality, higher ICU admission rates, and longer hospital stays ($p < 0.001$). Mortality increased with age, reaching 30.0% in patients aged ≥ 85 years ($p = 0.006$). Frailty scores (PRISMA-7 and TSFI) increased significantly with age, whereas ISS did not differ significantly across age groups. R [...]

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Mortality prediction in older adults with earthquake-related trauma: a model combining MEWS and laboratory biomarkers.

Ozen A, Acehan S, Satar S, Bulut EA, Gulen M, Nigiz K, Sevdembas S, Sonmez GO et al.

PURPOSE: This study aimed to identify laboratory parameters and clinical scores that can predict in-hospital mortality among patients aged 65 years and older who presented to the emergency department (ED) with earthquake-related trauma following the February 6 and 20, 2023 earthquakes.

METHODS: In this retrospective observational study, 362 patients with earthquake-related trauma between February 6-21, 2023, were included. Demographic characteristics, laboratory results, and clinical scores such as the Modified Early Warning Score (MEWS), Injury Severity Score (ISS), and Shock Index (SI) were analyzed.

RESULTS: Of the included patients, 53.9% ($n = 195$) were female, and the median age was 73 years (IQR: 69-80). The overall mortality rate was 9.4%. Multivariable logistic regression analysis identified MEWS (OR: 3.535; 95% CI: 2.397-5.212; $p < 0.001$), urea (OR: 1.009; 95% CI: 1.002-1.015; $p = 0.008$), and hs-Troponin I (OR: 1.001; 95% CI: 1.000-1.002; $p = 0.016$) as independent predictors of mortality. Receiver operating characteristic (ROC) analysis demonstrated an AUC of 0.936 for the combined model including MEWS, urea, and hs-Troponin I (95% CI: 0.875-0.996; $p < 0.001$). Among the variables, MEWS showed the highest AUC (0.903; 95% CI: 0.829-0.976). The optimal cutoff value for MEWS was determined as 1.5, with a sensitivity of 70.6% and a specificity of 97.6%.

CONCLUSIONS: In o [...]

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The economic burden in terms of cost of illness and generic health-related quality of life of posttraumatic long bone non-unions among the adult population of the Netherlands from a societal perspective.

van der Broeck LCA, Shchurov DG, Gabrio A, Lodewijks AJL, Poeze M, Blokhuis TJ, Evers S

PURPOSE: This study assessed the societal economic burden in terms of cost of illness and health-related quality of life (HRQoL) of posttraumatic long bone non-unions in the Netherlands.

METHODS: An incidence-based bottom-up approach was used, focusing on adult patients with posttraumatic long bone non-unions. The analysis included healthcare costs, patient and family costs, and productivity losses, measured using the iMCQ and iPCQ questionnaires. Cost evaluations followed Dutch costing guidelines, and productivity losses were calculated using the friction cost method. HRQoL was assessed with the EQ-5D-5L. A deterministic one-way sensitivity analysis varied baseline characteristics by $\pm 10\%$. Scenario analyses were conducted from healthcare and patient perspectives, as well as for patient subgroups.

RESULTS: Average costs per patient during the three months before the initial visit at a non-union clinic were €8,928 (healthcare, $N = 78$), €1,360 (patient and family, $N = 58$), and €4,313 (productivity losses, $N = 59$), the average of the observed data calculates to 10,831 ($N = 44$). The mean EQ-5D-5L utility score was 0.390 (± 0.29 SD). Subgroup analysis showed no significant cost increase for patients with infections or open fractures. However, a higher Non-Union Severity Score and a lower HRQoL were significantly associated with higher total costs.

CONCLUSION: Posttraumatic long [...]

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Impact of Preoperative Osteoporotic Medications on Total Hip Arthroplasty Outcomes.

Antonioli SS, Ruff G, Leung N, Patel A, Schwarzkopf R, Cohen-Rosenblum A

BACKGROUND: Osteoporosis (OP) is a common comorbidity in patients undergoing total hip arthroplasty (THA) and is a known risk factor for poor postoperative outcomes such as periprosthetic fracture (PPF). The impact of preoperative OP medications in patients with OP undergoing THA remains unclear. This study aimed to compare THA outcomes by OP diagnosis and preoperative bone strengthening medication usage.

METHODS: This was a retrospective review of primary elective THAs from June 2011-January 2024. Patients were stratified by OP diagnosis and OP medication usage, then propensity matched in a 1:2:3 ratio by age, sex, body mass index, and comorbidities. The resulting cohorts: (1) OP and medication usage for at least 1 year preoperatively and within 7 years of the procedure ($n = 296$), (2) OP and no medication usage ($n = 592$), and (3) no diagnosis of OP and no medication usage ($n = 888$) were then compared for postoperative outcomes.

RESULTS: Patients in cohort 1 were more likely to visit the emergency department ($P = .043$) or be readmitted for a surgical complication ($P = .007$) within 90 days. They were more likely to undergo posterior approach than cohorts 2 and 3 ($P = .016$), had the highest occurrence of cemented femoral stems (38.2%, $P < .001$), and had a higher incidence of revision for PPF (3.4%, vs 0.8% [cohort 2] and 1.2% [cohort 3], $P = .009$). Of the 12 revisions due to pe [...]

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